Open Source FEA using adapy

Generated with *adapy* v0.128.0.

This report describes the main verification work of the simulation results produced by the Open Source Finite Element (FE) solvers Code Aster and Calculix.

The motivation behind producing this report is to increase confidence in the conversion of FEM models in *adapy* and also the results from the various FEA solvers.

Introduction

The results from *Code Aster* and *Calculix* and tools will be compared with proprietary FEA solvers such as;

- Abaqus
- Sestra (part of DNV's Sesam suite)

The simulations are pre- and post processed using adapy and this report is automatically generated using paradoc as part of a ci/cd pipeline within the *adapy* repositories.

For each new addition of code affecting the handling of FEM in *adapy* a new set of analyses is performed, whereas the results are gathered in this auto-generated document.

Please note! This report will not go into detail into the specifics of why the results are different between different element representations. This report will only investigate and compare the results from different FEA software.

FEA Solvers

- Calculix v2.23
- Code Aster v18.1.7
- Abaqus v2025
- Sestra v11.3.1.0

Python packages

The following python packages were instrumental in the creation of this document and the FEA results herein.

Adapy

This report was generated with *adapy* v0.128.0.

The intention behind *adapy* is to make it easier to work with finite element models and BIM models and act as a high-level scripting library for parametric and procedural modelling techniques.

Paradoc

paradoc was created to simplify the generation of reports by creating the structure of the document and text in markdown with a string substitution scheme that lets you easily pass in tables, functions and finally be able to produce production ready documents in PDF, ODT or Microsoft Word (docx).

Summary

A summary of all performed analyses will be performed in this section and a percentage difference for each FEA solver per type of analysis will be calculated and presented.

Analyses

Eigenvalue Analysis

A simple cantilevered beam is subjected to eigenvalue analysis with a total of 11 eigenmodes requested. 49 cases are compared across solvers: abaqus, calculix, code_aster, sesam.

The beam model will be analyzed by varying the following parameters:

- Mesh type (Solid, Shell, Line)
- Mesh element types (TET, HEX, TRI, QUAD)
- Mesh element order (1st and 2nd order)
- Reduced integration elements (R)
- Different FEA solvers (Calculix, Code Aster, Abaqus, Sesam)

Model Description

Property	Value
Description	Beam(“MyBeam”, [0.0, 0.5, 0.5], [3.0, 0.5, 0.5], “IPE400”, “S420”)
Length	3.00 m
Section	IPE400
Material	S420
Young’s modulus	2.100e+11 Pa
Yield stress	4.200e+08 Pa
Density	7850 kg/m³

Cantilever beam geometry.

Figure 1: Cantilever beam geometry.

Summary

This section presents a comparison between the calculated results from the eigenvalue analysis of a cantilever beam using the different FEA tools.

Note! The `Sestra` and `Abaqus` analyses are not performed as part of the github actions step. The eigenmode results are kept in cached json files and imported during the compilation of the report using **paradoc**.

Eigenfrequency vs. mode number

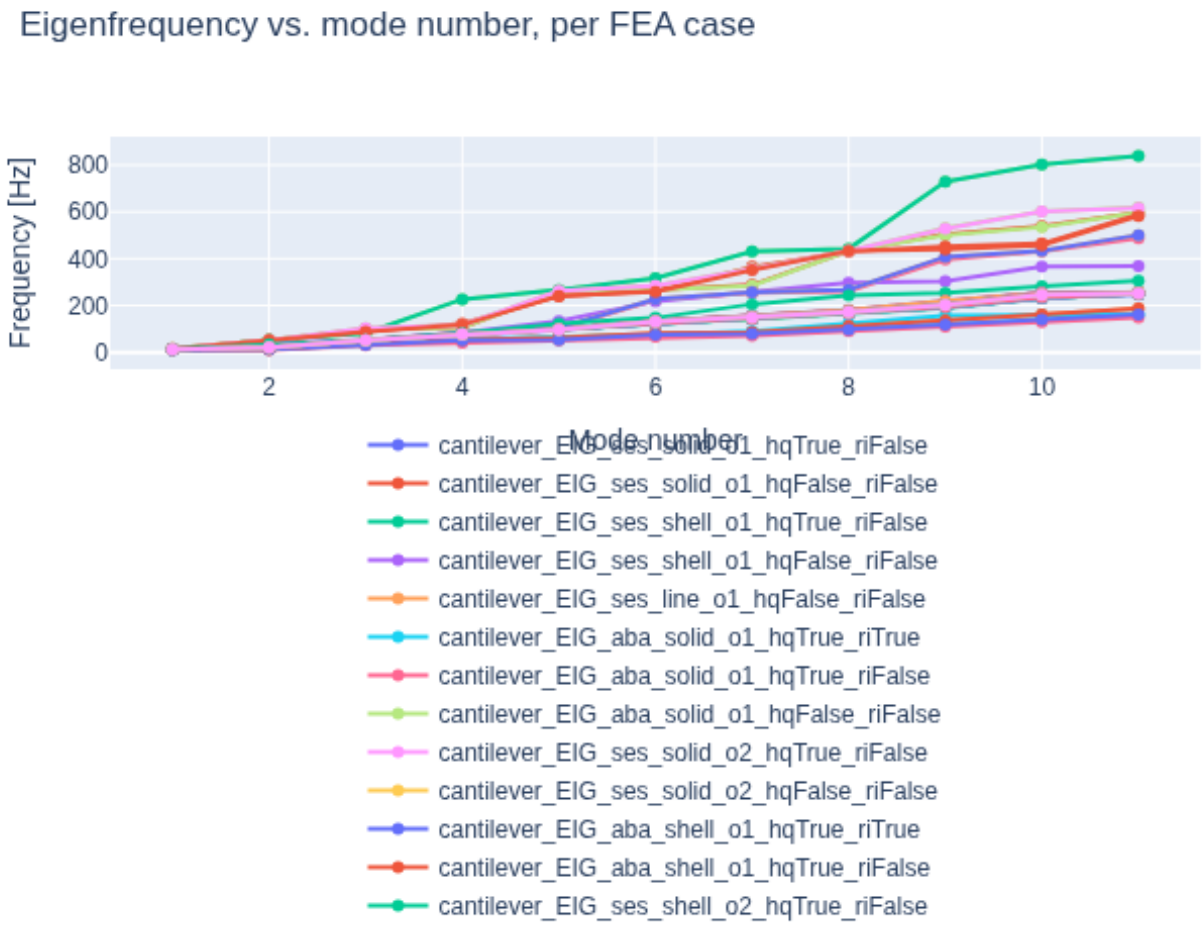


Figure 2: Eigenfrequency vs. mode number across FEA cases.

Comparison tables

Columns are sortable; the table grammar `{tbl:sortby:Mode:asc;index:no}` sets the initial sort and hides the index column.

Table 2: Eigenfrequency comparison (Hz) — solid, 1st order.

Mode	ses_HEX	ses_TET	aba_HEXR	aba_HEX	aba_TET	ccx_TET	ca_TET	ccx_HEX	ca_HEX	ccx_HEXR	ca_HEXR
1	12.9766	15.7932	12.514	13.435	15.79	15.9285	15.9277	13.6205	13.6205	14.3018	13.5291
2	20.3015	51.5418	34.157	20.731	51.532	51.5896	51.5872	21.0685	21.0685	51.6653	34.7401
3	50.6181	97.7314	49.598	50.678	97.588	102.446	102.41	50.8252	50.8252	88.2331	50.7365
4	77.3921	105.691	55.271	82.799	104.8	116.744	116.512	84.066	84.066	116.873	82.2962
5	98.1185	260.887	64.305	100.19	260.59	261.877	261.806	102.334	102.334	240.766	123.303
6	133.016	267.131	74.866	223.14	266.26	284.697	284.465	227.25	227.25	258.286	147.139
7	153.833	287.461	90.033	254.98	285.05	357.062	356.306	255.872	255.872	351.823	205.267
8	178.565	432.347	124.58	260.1	432.31	432.36	432.35	266.519	266.519	431.955	243.985
9	214.689	503.862	155.99	396.57	500.86	529.297	528.489	407.869	407.869	439.038	254.838
10	253.463	539.506	159.95	431.8	534.26	601.943	601.586	432.207	432.207	456.762	282.888
11	256.05	598.2	162.56	486.96	596.78	617.646	616.155	500.931	500.931	582.562	305.247

Table 3: Eigenfrequency comparison (Hz) — solid, 2nd order.

Mode	ses_HEX	ses_TET	aba_HEXR	aba_HEX	aba_TET	ccx_TET	ca_TET	ccx_HEX	ca_HEX	ccx_HEXR	ca_HEXR
1	12.9541	12.9537	12.943	12.955	12.954	12.9542	12.9542	12.9551	12.9551	12.9428	12.9427
2	20.0445	20.2945	20.029	20.15	20.294	20.2876	20.2877	20.1503	20.1503	20.0288	20.0288
3	50.5342	50.5401	50.498	50.541	50.54	50.5401	50.5402	50.5412	50.5412	50.4977	50.4977
4	76.7467	77.1278	76.687	76.89	77.128	77.2297	77.2297	76.8903	76.8903	76.6876	76.6875
5	97.4723	97.8296	97.391	97.594	97.83	97.8506	97.8505	97.5939	97.5939	97.3913	97.3912
6	130.865	132.478	130.81	131.37	132.48	133.015	133.014	131.365	131.365	130.814	130.809

Mode	ses_HEX	ses_TET	aba_HEXR	aba_HEX	aba_TET	ccx_TET	ca_TET	ccx_HEX	ca_HEX	ccx_HEXR	ca_HEXR
7	148.496	153.631	148.47	149.96	153.63	154.804	154.802	149.958	149.958	148.478	148.474
8	172.243	178.114	172.17	173.92	178.12	181.029	181.026	173.925	173.925	172.177	172.171
9	201.904	215.815	201.84	205.29	215.82	219.501	219.497	205.292	205.292	201.846	201.839
10	246.353	253.203	246.17	251.35	253.2	253.193	253.193	251.347	251.347	246.177	246.165
11	252.742	254.466	252.57	252.92	254.47	254.685	254.684	252.917	252.917	252.573	252.573

Table 4: Eigenfrequency comparison (Hz) — shell, 1st order.

Mode	ses_QUAD	ses_TRI	aba_QUADR	aba_QUAD	aba_TRIR	ccx_TRI	ca_TRI	ccx_QUAD	ca_QUAD	ccx_QUADR
1	12.7628	13.6414	12.373	12.78	13.6869	13.7254	13.642	12.7639	13.1463	14.2558
2	19.3943	20.1983	19.23	19.593	20.3643	36.6896	20.4181	19.6304	19.9719	53.2618
3	51.9912	52.1922	51.849	51.998	52.1874	52.2485	52.1927	51.9997	52.0595	87.9153
4	75.1918	79.5226	73.558	75.683	80.2058	83.7093	79.6596	75.4574	77.2361	121.85
5	95.9297	101.756	93.709	96.393	102.043	132.928	102.01	96.2959	98.7619	239.804
6	125.116	125.093	126.87	126.96	124.093	218.167	125.459	125.898	125.442	262.83
7	141.556	143.164	144.58	145.57	145.486	259.245	144.895	145.282	143.864	365.1
8	165.03	166.668	168.38	169.39	168.62	297.856	168.951	169.441	167.974	431.774
9	190.099	191.533	197.51	198.75	199.719	304.464	195.184	200.494	193.96	451.745
10	229.46	230.489	241.27	242.55	242.995	365.943	235.115	247.276	233.817	464.44
11	250.786	257.362	245.07	252.49	257.447	368.764	257.409	251.554	257.105	589.24

Table 5: Eigenfrequency comparison (Hz) — shell, 2nd order.

Mode	ses_QUAD	ses_TRI	aba_QUADR	aba_QUAD	aba_TRI	ccx_TRI	ca_TRI	ccx_QUAD	ca_QUAD	ccx_QUADR
1	12.7554	12.7558	12.758	12.758	12.756	12.7562	12.7566	12.7576	12.7561	12.7559
2	19.4731	19.5048	19.486	19.486	19.567	19.5254	19.4509	19.5638	19.4736	19.5021
3	51.9702	51.9704	51.998	51.998	51.969	51.974	51.97	51.9801	51.97	51.9741
4	75.1245	75.1716	75.149	75.149	75.145	75.1848	75.1443	75.2685	75.1252	75.1399
5	95.8886	95.9337	95.92	95.92	96.001	95.9713	95.8707	96.0372	95.8893	95.9349
6	124.987	125.173	125	125	125.21	125.465	125.087	125.889	124.987	125.257
7	142.234	142.768	142.32	142.32	142.77	142.995	142.359	143.72	142.233	142.497
8	165.718	166.334	165.85	165.85	166.6	166.579	165.79	167.347	165.719	165.981
9	191.999	193.308	192.18	192.18	192.84	194.014	192.406	195.577	191.998	192.87
10	232.626	234.613	232.87	232.87	233.36	236.042	233.389	238.389	232.628	234.347
11	250.095	250.24	250.17	250.17	250.13	250.287	250.175	250.521	250.096	250.147

Table 6: Eigenfrequency comparison (Hz) — line, 1st order.

Mode	ses	aba	ccx	ca	ccxR
1	12.2454	10.785	12.9734	12.2454	12.9734
2	12.9431	12.928	52.5867	12.9793	52.5867
3	36.7525	32.342	81.1511	36.7525	81.1511
4	50.5542	50.543	226.646	52.9559	226.646
5	61.3087	53.855	267.32	61.3087	267.32
6	79.793	75.296	316.915	81.3401	316.915
7	85.9466	79.59	431.04	85.9466	431.04
8	110.699	96.637	442.61	110.699	442.61
9	135.6	117.85	728.596	135.6	728.596
10	160.681	138.9	802.316	160.681	802.316
11	185.976	159.77	838.75	185.976	838.75

Table 7: Eigenfrequency comparison (Hz) — line, 2nd order.

Mode	ses	aba
1	10.039	10.786
2	10.6144	12.932
3	30.1078	32.358
4	41.2358	50.557
5	50.1488	53.93
6	65.3265	75.502
7	70.1429	79.684
8	90.0695	97.074
9	109.907	118.65
10	129.631	140.22

Mode	ses	aba
11	149.217	161.8

Effective modal mass

Effective modal mass [kg] per case, summed over the captured modes in the global X / Y / Z directions. Every solver reports it; Code Aster and Sesam give translational mass only. Summing over enough modes approaches the structure's total mass in each direction.

Table 8: Effective modal mass [kg] per case, summed over the captured modes in the global X/Y/Z directions. Code_Aster and Sesam report translational effective mass only.

Case	Modes	ΣMeff X	ΣMeff Y	ΣMeff Z
aba_line_o1	11	0	152.8	118.2
aba_line_o2	11	0	152.8	118.2
aba_shell_o1_QUAD	11	0	168	122
aba_shell_o1_QUADR	11	0	168.4	122
aba_shell_o2_QUAD	11	0	167.8	122
aba_shell_o2_QUADR	11	0	167.8	122
aba_shell_o2_TRI	11	0	167.9	122
aba_solid_o1_HEX	11	153.5	170.5	158.2
aba_solid_o1_HEXR	11	0	154.3	119
aba_solid_o1_TET	11	153.4	172	170.9
aba_solid_o2_HEX	11	0	163.3	158.6
aba_solid_o2_HEXR	11	0	163.2	158.7
aba_solid_o2_TET	11	0	162	158.6
ca_line_o1	11	0	152.3	116.5
ca_shell_o1_QUAD	11	0	167.6	163.7
ca_shell_o1_TRI	11	0	167.3	163.7
ca_shell_o2_QUAD	11	0	167.8	122
ca_shell_o2_TRI	11	0	167.8	122
ca_solid_o1_HEX	11	153.5	170.8	158.3
ca_solid_o1_HEXR	11	0	164.7	158.5
ca_solid_o1_TET	11	153.4	172.6	171
ca_solid_o2_HEX	11	0	163.3	158.6
ca_solid_o2_HEXR	11	0	163.2	158.7
ca_solid_o2_TET	11	0	162.2	158.6
ccx_line_o1	11	153.9	174.4	161.2
ccx_line_o1R	11	153.9	174.4	161.2
ccx_shell_o1_QUAD	11	0	168.1	122
ccx_shell_o1_QUADR	11	158	176.8	176.5
ccx_shell_o1_TRI	11	0	174.8	163.6
ccx_shell_o2_QUAD	11	0	167.8	122
ccx_shell_o2_QUADR	11	0	167.8	122
ccx_shell_o2_TRI	11	0	167.8	122
ccx_solid_o1_HEX	11	153.4	170.7	158.2
ccx_solid_o1_HEXR	11	153.5	171.8	171.3
ccx_solid_o1_TET	11	153.4	172.6	171
ccx_solid_o2_HEX	11	0	163.3	158.6
ccx_solid_o2_HEXR	11	0	163.2	158.7
ccx_solid_o2_TET	11	0	162.2	158.6
ses_line_o1	11	0	153	120.2
ses_line_o2	11	0	231.8	180
ses_shell_o1_QUAD	11	0	167.6	122
ses_shell_o1_TRI	11	0	167.1	163.6
ses_shell_o2_QUAD	11	0	167.8	122
ses_shell_o2_TRI	11	0	167.9	122
ses_solid_o1_HEX	11	0	161.7	158.6
ses_solid_o1_TET	11	153.4	172.1	170.9
ses_solid_o2_HEX	11	0	163.1	158.7
ses_solid_o2_TET	11	0	162	158.6

The per-mode effective masses and participation factors, compared across solvers for every mesh configuration, are in the appendix.

Short description:

Fem formats

- ccx: Calculix

- ca: Code Aster
- ses: Sesam
- aba: Abaqus

Element Types

- TET: Tetrahedrons
- HEX: Hexahedrons
- TRI: Triangle
- QUAD: Quadrilateral

Reduced Integration

All element types ending with capital R (QUAD**R**, HEX**R** etc.) are reduced integration elements.

Nonlinear Analysis

A simple cantilever beam with a series of cutouts is subjugated to a large vertical acceleration field. The level of plastic strains are monitored for comparison across different FE tools.

Model Description

Summary

Plate Analysis

The analyses above are one beam, in three representations. This section is the plate case: an `ada.Plate` object, meshed into shell elements by adapy's own meshing module, written out as an input deck per solver, and checked against a closed form as well as against the other solvers.

Nothing here drives a solver through its API. Each solver reads a file adapy wrote — an `.inp`, a `.fem`, a `.comm`/`.med` — which is the same route a user's own model takes, so what this section verifies is the thing adapy actually ships.

8 cases ran on the machine that built this report, using: calculix, code_aster. The remaining columns come from cached results measured elsewhere, the same arrangement the eigenvalue section uses for Abaqus and Sestra.

Model Description

A rectangular strip, simply supported on its two short edges and held in cylindrical bending on the two long ones, under a uniform surface pressure.

Property	Value
Span	4.00 m
Width	0.50 m
Thickness	0.010 m
Pressure	1000 Pa
Material	S355 (E = 210 GPa, ν = 0.3, ρ = 7850 kg/m^3)

Plate strip geometry.

Figure 3: Plate strip geometry.

Plate strip with a T-profile stiffener on its centreline.

Figure 4: Plate strip with a T-profile stiffener on its centreline.

The span-to-thickness ratio is 400, so transverse shear sits several orders below the discretisation error and the thin-plate closed forms are the right reference rather than an approximation of one.

The supports are written as four node sets rather than one per edge. Simple support and cylindrical bending overlap at the four corners, and the obvious spelling constrains u2 twice there: Abaqus and Sestra accept that, while Code_Aster refuses it outright (`<ASSEMBLA_26> … est bloqué plusieurs fois`). A model that only some solvers will read is not a cross-solver case, so the corner nodes carry only the rotation the long edges add.

Static analysis under pressure

The static case is the only one that reaches the distributed-load records — Abaqus' `*Dsload`, Calculix' `*DLOAD`, Code_Aster's `FORCE_COQUE` and Sesam's `BEUSLO`. The eigenvalue analyses reach the mass and stiffness matrices and nothing else, so a pressure written wrongly is invisible to them.

Mid-span deflection against the closed form $5\,q\,L^4 / (384\,D)$, with $D = E\,t^3 / (12\,(1-\nu^2))$ — which evaluates to 0.173333 m for this strip. Halving the seed should quarter the remaining error; the three rows are a factor of two apart so the order is readable off the table rather than asserted.

Table 10: Plate strip mid-span deflection [m] under 1 kPa, by solver and mesh seed. The closed form is 5 q L^4 / (384 D) for a strip in cylindrical bending.

Seed [m]	aba	ca	ccx	ses	closed form
0.125	0.173065	0.173198	0.173065	0.173198	0.173333
0.0625	0.173269	0.173299	0.173268	0.173299	0.173333
0.03125	0.173319	0.173325	0.173319	0.173325	0.173333

The same strip with a T-profile stiffener along its centreline, meshed as line elements sharing the shells' nodes. There is no closed form for it — a parallel-spring estimate is not one — so this is an agreement case between solvers, and it is what carries a T section through the deck writers.

Table 11: The same strip with a T-profile stiffener on its centreline [m]. No closed form: this is a cross-solver agreement case.

Seed [m]	aba_st	ca_st	ses_st
0.125	0.000584531	0.000710295	0.000720708
0.0625	0.000585478	0.000710711	0.000720976
0.03125	0.000585996	0.000710815	0.000720795

Sign convention

Every writer above places a positive pressure so that it pushes **into** the face it names. That is Abaqus' convention, and it has to be shared, because a load direction that depends on the output format makes a cross-solver comparison meaningless.

It is worth stating which face that is, because it is not a global direction: gmsh winds every element of this strip with its normal along −z, so the positive face looks *down* and a positive pressure on it moves the strip *up*. The deflections above are therefore positive. `tests/fem/test_pressure_load_cross_format.py` pins this per format.

Eigenvalue analysis

The same plate, as an eigenvalue analysis, against $f_n = n^2\,\pi / (2\,L^2)\,\sqrt{D / (\rho\,t)}$ — the simply supported beam frequencies with the beam's $EI / (\rho\,A)$ replaced by the strip's $D / (\rho\,t)$. Valid for the cylindrical modes, which are the ones the long-edge constraint admits.

Table 12: Plate strip eigenfrequencies [Hz], by solver, at a 0.0625 m seed. The closed form is f_n = n^2 pi / (2 L^2) sqrt(D / (rho t)), valid for the cylindrical modes the long-edge constraint admits.

Mode	aba	ca	ccx	ses	closed form
1	1.5371	1.53661	1.53675	1.53661	1.53661
2	6.1536	6.14644	6.14861	6.14644	6.14644
3	13.866	13.8295	13.8405	13.8295	13.8295
4	24.7	24.5858	24.6205	24.5858	24.5857
5	38.695	38.4153	38.5002	38.4153	38.4152
6	55.9	55.3181	55.4943	55.3182	55.3179

Short description of the solver column names:

- ccx: Calculix
- ca: Code Aster
- ses: Sesam
- aba: Abaqus

A `_st` suffix marks the stiffened variant.

Detailed results

Mode shapes and deflected shapes from every format that ran on the machine that built this report, laid out so the formats can be read against each other: one section per mesh configuration, one heading per mode, and the formats side by side under it. A swapped mode pair or a differently deflected shape shows up as two pictures that do not match.

Abaqus and Sesam cannot run where this report is built. Their mode shapes are the raw results a licensed machine committed beside its cached frequencies, and the figures are rendered from them here; a case with no committed results has no figures. Solver versions: Abaqus v2025, Calculix v2.23, Code Aster v18.1.7, Sesam v11.3.1.0.

Cantilever modal participation

Per-mode effective modal mass and participation factors in the global X / Y / Z directions, for every mesh configuration, with the formats side by side. Code Aster and Sesam report the translational directions only.

A participation factor Γ is relative to how the solver normalised its eigenvectors: Abaqus scales the largest displacement to 1, the others scale to unit generalised mass. Raw factors from different solvers therefore do not compare. The tables give the factor scaled to unit generalised mass, sign(Γ)·√Meff, which is what the others report as-is. Its sign is the solver's choice of eigenvector sign and may differ from mode to mode. The effective mass Meff = $\Gamma^2 \cdot$m_gen does not depend on the normalisation.

Solid, 1st order

Table 13: Effective modal mass [kg] per mode, global X/Y/Z — solid, order 1.

Mode	Direction	ses_HEX	ses_TET	aba_HEXR	aba_HEX	aba_TET	ccx_TET	ca_TET	ccx_HEX	ca_HEX	ccx_HEXR	ca_HEXR
1	X	0	0	0	0	0	0	0	0	0	0	0
1	Y	116.4	117.5	116.9	116.4	117.5	118.2	118.2	116.4	116.4	116.5	116.4
1	Z	0	0	0	0	0	0	0	0	0	0	0
2	X	0	0	0	0	0	0	0	0	0	0	0
2	Y	0	0	0	0	0	0	0	0	0	0	0
2	Z	0	118.3	0	0	118.2	118.4	118.4	0	0	118.5	0
3	X	0	0	0	0	0	0	0	0	0	0	0
3	Y	0	34.9	0	0	34.9	35.5	35.5	0	0	36.2	0
3	Z	118.3	0	119	118.2	0	0	0	118.2	118.2	0	118.3
4	X	0	0	0	0	0	0	0	0	0	0	0
4	Y	35.9	0	2.1	36	0	0	0	36	36.1	0	36.1
4	Z	0	0	0	0	0	0	0	0	0	0	0

Mode	Direction	ses_HEX	ses_TET	aba_HEXR	aba_HEX	aba_TET	ccx_TET	ca_TET	ccx_HEX	ca_HEX	ccx_HEXR	ca_HEXR
5	X	0	0	0	0	0	0	0	0	0	0	0
5	Y	0	0	0	0	0	0	0	0	0	12.6	0
5	Z	0	39.8	0	0	39.7	39.9	39.9	0	0	0	0
6	X	0	0	0	0	0	0	0	0	0	0	0
6	Y	0	13.3	31.4	12.4	13.2	12.8	12.8	12.5	12.5	0	0
6	Z	0	0	0	0	0.1	0	0	0	0	40.2	0
7	X	0	0	0	0	0	0	0	0	0	0	0
7	Y	3.2	0	2.8	0	0	0	0	0	0	0	8.9
7	Z	0	0	0	40	0	0	0	40	40.1	0	0
8	X	0	153.4	0	0	153.4	153.4	153.4	0	0	153.5	0
8	Y	5.7	0	0	0	0	0	0	0	0	0	3.1
8	Z	0	0	0	0	0	0	0	0	0	0	0.3
9	X	0	0	0	0	0	0	0	0	0	0	0
9	Y	0.4	6.3	1.1	5.7	6.3	6.1	6.1	5.9	5.9	0.4	0
9	Z	0	0	0	0	0	0	0	0	0	0	39.9
10	X	0	0	0	153.5	0	0	0	153.4	153.5	0	0
10	Y	0	0	0	0	0	0	0	0	0	6.1	0
10	Z	40.3	0	0	0	0	12.8	12.8	0	0	0	0
11	X	0	0	0	0	0	0	0	0	0	0	0
11	Y	0	0	0	0	0	0	0	0	0	0	0
11	Z	0	12.8	0	0	12.8	0	0	0	0	12.7	0

Table 14: Participation factor per mode scaled to unit generalised mass [√kg], global X/Y/Z — solid, order 1.

Mode	Direction	ses_HEX	ses_TET	aba_HEXR	aba_HEX	aba_TET	ccx_TET	ca_TET	ccx_HEX	ca_HEX	ccx_HEXR	ca_HEXR
1	X	-0	0	0	-0	0	-0	0	-0	0	0	0
1	Y	10.79	-10.841	10.813	10.788	10.838	-10.871	10.871	10.787	10.787	10.794	10.791
1	Z	0	0.017	-0	-0	-0.017	0.019	-0.019	-0	-0	0	-0.001
2	X	-0	-0	0	0	0	0	0	-0	-0	-0	0.001
2	Y	-0	-0.028	0	0	0.028	0.03	0.03	0	0	0	0.022
2	Z	0	-10.876	-0	-0	10.874	10.88	10.88	-0	0	-10.884	-0
3	X	-0	0	-0	-0	-0	0	-0	-0	-0	0	0
3	Y	-0	5.911	0	0	-5.909	5.961	-5.961	0	0	-6.013	0
3	Z	10.876	-0.019	10.907	10.872	0.019	-0.02	0.02	10.871	10.872	-0	10.876
4	X	-0	-0.003	-0	-0	0.003	0.003	0.003	0	0	0	-0
4	Y	5.995	0.069	-1.456	-6.002	-0.075	-0.108	-0.11	6.004	-6.004	-0	-6.01
4	Z	-0	-0.003	0	0	0.003	0.007	0.008	0	-0	-0	-0.004
5	X	0	-0.001	-0	0	-0.001	0	-0	-0	0	-0	-0.002
5	Y	-0	0.122	-0	-0	0.132	-0.028	0.029	0	-0	-3.549	-0.084
5	Z	0	-6.305	0	0	-6.305	6.313	-6.315	-0	-0	0	0.006
6	X	-0	0	0	-0	0	0	-0.001	0	-0	0	-0.001
6	Y	-0.114	3.642	-5.603	3.522	3.638	-3.576	3.577	3.53	3.532	-0	0.192
6	Z	0	0.207	-0	0	0.226	-0.056	0.057	0	-0	-6.34	-0.015
7	X	-0	0	-0	-0	0	-0.021	-0.021	0	0	0	0
7	Y	-1.788	0.027	-1.671	0	0.03	0.04	0.04	-0	-0	-0	2.99
7	Z	0	-0.033	0	-6.326	-0.035	-0.02	-0.02	6.327	-6.329	-0	0.071
8	X	0	12.384	-0	-0	12.385	12.384	12.386	0	-0	12.389	0.001
8	Y	2.39	-0.001	-0	0	-0.001	0	0	0	0	0	1.76
8	Z	0	0	-0	0	0	-0	-0	-0	-0	0	-0.503
9	X	-0	-0.005	-0	0	-0.006	0.001	0.001	-0	-0	0	0.001
9	Y	0.662	-2.518	-1.056	-2.385	-2.515	-2.462	-2.464	2.426	-2.43	-0.619	-0.115
9	Z	-0	0.007	-0	0	0.007	0.064	0.064	0	0	-0	-6.318
10	X	-0	-0.015	0.019	12.391	-0.016	0.001	-0.001	-12.385	12.388	-0	0.006
10	Y	0	-0.015	-0	0	-0.017	-0.045	0.045	-0	-0	2.478	0.086
10	Z	-6.346	0.003	0	-0	0.003	-3.572	3.576	0	0	-0	-0.006
11	X	0	0.001	-0	0	-0.001	-0.012	0.012	0	0	-0	0.005
11	Y	-0	0	0	0	-0	0.016	-0.016	-0	0	-0	0.009
11	Z	0	-3.578	-0	0	3.578	0.029	-0.031	0	-0	3.558	-0.001

Solid, 2nd order

Table 15: Effective modal mass [kg] per mode, global X/Y/Z — solid, order 2.

Mode	Direction	ses_HEX	ses_TET	aba_HEXR	aba_HEX	aba_TET	ccx_TET	ca_TET	ccx_HEX	ca_HEX	ccx_HEXR	ca_HEXR
1	X	0	0	0	0	0	0	0	0	0	0	0
1	Y	116.5	116.5	116.6	116.5	116.5	116.5	116.5	116.5	116.5	116.6	116.6
1	Z	0	0	0	0	0	0	0	0	0	0	0

Mode	Direction	ses_HEX	ses_TET	aba_HEXR	aba_HEX	aba_TET	ccx_TET	ca_TET	ccx_HEX	ca_HEX	ccx_HEXR	ca_HEXR
2	X	0	0	0	0	0	0	0	0	0	0	0
2	Y	0	0	0	0	0	0	0	0	0	0	0
2	Z	0	0	0	0	0	0	0	0	0	0	0
3	X	0	0	0	0	0	0	0	0	0	0	0
3	Y	0	0	0	0	0	0	0	0	0	0	0
3	Z	118.4	118.4	118.4	118.4	118.4	118.4	118.4	118.4	118.4	118.4	118.4
4	X	0	0	0	0	0	0	0	0	0	0	0
4	Y	35.9	36	35.9	35.9	36	36	36	35.9	35.9	35.9	35.9
4	Z	0	0	0	0	0	0	0	0	0	0	0
5	X	0	0	0	0	0	0	0	0	0	0	0
5	Y	0	0	0	0	0	0	0	0	0	0	0
5	Z	0	0	0	0	0	0	0	0	0	0	0
6	X	0	0	0	0	0	0	0	0	0	0	0
6	Y	0	0	0	0	0	0	0	0	0	0	0
6	Z	0	0	0	0	0	0	0	0	0	0	0
7	X	0	0	0	0	0	0	0	0	0	0	0
7	Y	3.4	3.1	3.4	3.3	3.1	3.2	3.2	3.3	3.3	3.4	3.4
7	Z	0	0	0	0	0	0	0	0	0	0	0
8	X	0	0	0	0	0	0	0	0	0	0	0
8	Y	5.3	5.8	5.3	5.4	5.8	5.9	5.9	5.4	5.4	5.3	5.3
8	Z	0	0	0	0	0	0	0	0	0	0	0
9	X	0	0	0	0	0	0	0	0	0	0	0
9	Y	0.6	0.6	0.6	0.5	0.6	0.5	0.5	0.5	0.5	0.6	0.6
9	Z	0	0	0	0	0	0	0	0	0	0	0
10	X	0	0	0	0	0	0	0	0	0	0	0
10	Y	1.5	0	1.5	1.6	0	0	0	1.6	1.6	1.5	1.5
10	Z	0	40.2	0	0	40.2	40.2	40.2	0	0	0	0
11	X	0	0	0	0	0	0	0	0	0	0	0
11	Y	0	0	0	0	0	0	0	0	0	0	0
11	Z	40.3	0	40.3	40.3	0	0	0	40.3	40.3	40.3	40.3

Table 16: Participation factor per mode scaled to unit generalised mass [√kg], global X/Y/Z — solid, order 2.

Mode	Direction	ses_HEX	ses_TET	aba_HEXR	aba_HEX	aba_TET	ccx_TET	ca_TET	ccx_HEX	ca_HEX	ccx_HEXR	ca_HEXR
1	X	0	-0	-0	0	0	-0	0	-0	-0	0	0
1	Y	10.794	-10.794	10.796	10.794	10.794	-10.794	10.794	10.794	10.794	10.796	10.796
1	Z	-0	-0	0	0	0	-0	0	-0	0	0	-0
2	X	0	0	-0	-0	0	-0	0	-0	-0	-0	0
2	Y	-0	-0	0	0	-0	-0	0	0	0	-0	0
2	Z	0	0	-0	0	0	-0	0	0	0	0	-0
3	X	0	0	-0	-0	0	-0	-0	-0	-0	-0	-0
3	Y	0	-0	-0	-0	-0	-0	-0	0	-0	-0	0
3	Z	10.88	10.879	10.882	10.879	10.879	10.879	10.879	10.879	10.879	10.882	10.882
4	X	-0	0	-0	-0	0	-0	-0	0	0	0	-0
4	Y	-5.991	-5.997	-5.992	-5.993	-5.997	-6.004	-6.004	-5.993	-5.993	-5.992	-5.992
4	Z	0	-0	-0	-0	-0	0	0	0	-0	-0	-0
5	X	-0	0	0	0	-0	-0	0	0	-0	-0	-0
5	Y	0	0.001	-0	-0	-0.001	0.001	-0.001	0	-0	-0	-0
5	Z	-0	-0	-0	-0	0	-0	0	-0	0	0	0
6	X	0	-0	-0	0	0	0	0	0	-0	-0	0
6	Y	-0.096	-0.119	-0.097	-0.103	0.119	-0.093	-0.093	0.103	-0.103	0.096	-0.097
6	Z	0	0	0	-0	-0	0	0	0	-0	0	-0
7	X	0	0	-0	0	-0	0	0	-0	0	0	0
7	Y	-1.839	-1.768	1.84	1.818	1.768	1.799	1.799	1.818	1.818	1.84	1.84
7	Z	0	0	-0	0	-0	0	0	0	0	0	-0
8	X	-0	0	0	0	-0	0	0	-0	-0	0	0
8	Y	-2.305	-2.406	2.304	2.328	2.405	2.436	2.435	2.328	2.328	2.304	2.304
8	Z	-0	0	0	-0	-0	-0	-0	-0	-0	0	-0
9	X	-0	0	-0	0	-0	0	0	-0	0	-0	-0
9	Y	-0.746	-0.774	-0.747	-0.719	0.774	0.69	0.69	0.72	-0.719	0.747	-0.747
9	Z	0	-0	0	0	0	-0	-0	0	-0	0	-0
10	X	-0	-0	0	0	-0	-0	0	-0	-0	0	0
10	Y	1.211	0	1.212	1.267	0	-0	0	-1.267	1.267	-1.213	1.212
10	Z	0	-6.344	-0	-0	-6.344	6.344	-6.344	0	-0	0	-0
11	X	-0	-0	0	0	-0	-0	0	-0	-0	0	-0
11	Y	0	-0.001	0	0	-0.001	-0.008	0.008	0	-0	-0	-0
11	Z	6.346	-0.006	-6.345	-6.344	-0.006	0.004	-0.004	6.345	-6.344	6.345	-6.345

Shell, 1st order

Table 17: Effective modal mass [kg] per mode, global X/Y/Z — shell, order 1.

Mode	Direction	ses_QUAD	ses_TRI	aba_QUADR	aba_QUAD	ccx_TRI	ca_TRI	ccx_QUAD	ca_QUAD	ccx_QUADR
1	X	0	0	0	0	0	0	0	0	0
1	Y	119.9	119.9	120	119.9	119.7	119.9	119.9	119.9	119.9
1	Z	0	0	0	0	0	0	0	0	0
2	X	0	0	0	0	0	0	0	0	0
2	Y	0	0	0	0	0	0	0	0	0
2	Z	0	0	0	0	0	0	0	0	122.1
3	X	0	0	0	0	0	0	0	0	0
3	Y	0	0	0	0	0	0	0	0	37.2
3	Z	122	122	122	122	122.1	122	122	122	0
4	X	0	0	0	0	0	0	0	0	0
4	Y	36.8	36.6	37	36.9	37.3	36.6	36.9	36.8	0
4	Z	0	0	0	0	0	0	0	0	0
5	X	0	0	0	0	0	0	0	0	0
5	Y	0	0	0	0	0	0	0	0	13
5	Z	0	0	0	0	0	0	0	0	0
6	X	0	0	0	0	0	0	0	0	0
6	Y	0	0.1	0	0	12.7	0.1	0	0	0
6	Z	0	0	0	0	0	0	0	0	41.5
7	X	0	0	0	0	0	0	0	0	0
7	Y	3.5	2.7	3.8	3.3	0	2.7	3.3	3.2	0
7	Z	0	0	0	0	41.6	0	0	0	0
8	X	0	0	0	0	0	0	0	0	158
8	Y	5.4	5.9	5.4	5.7	0	6	5.8	5.7	0
8	Z	0	0	0	0	0	0	0	0	0
9	X	0	0	0	0	0	0	0	0	0
9	Y	0.6	0.6	0.6	0.6	0	0.6	0.6	0.6	5.7
9	Z	0	0	0	0	0	0	0	0	0
10	X	0	0	0	0	0	0	0	0	0
10	Y	1.3	1.3	1.6	1.5	1.5	1.3	1.7	1.3	1
10	Z	0	0	0	0	0	0	0	0	0
11	X	0	0	0	0	0	0	0	0	0
11	Y	0	0	0	0	3.6	0	0	0	0
11	Z	0	41.6	0	0	0	41.6	0	41.6	12.9

Table 18: Participation factor per mode scaled to unit generalised mass [√kg], global X/Y/Z — shell, order 1.

Mode	Direction	ses_QUAD	ses_TRI	aba_QUADR	aba_QUAD	ccx_TRI	ca_TRI	ccx_QUAD	ca_QUAD	ccx_QUADR
1	X	-0	-0	-0	-0	0	-0	0	0	-0
1	Y	10.95	-10.951	10.952	10.951	10.941	10.951	10.95	10.951	10.95
1	Z	-0	-0	-0	-0	0	0	0	-0	0
2	X	-0	0	-0	-0	-0	0	0	-0	-0
2	Y	-0	-0	0	0	0.002	-0	0	-0	0
2	Z	-0	0	0	0	-0	0	0	0	-11.051
3	X	0	-0	-0	-0	-0	0	-0	-0	-0
3	Y	-0	0	0	0	0	-0	0	0	6.101
3	Z	-11.046	-11.047	11.046	11.046	-11.048	11.047	-11.045	11.046	0
4	X	-0	0	-0	-0	-0	0	-0	-0	0
4	Y	-6.067	-6.048	-6.081	-6.076	-6.104	-6.053	-6.072	6.065	0
4	Z	-0	-0	-0	-0	0	-0	0	0	0
5	X	0	-0	0	-0	0	0	-0	-0	-0
5	Y	0	0	-0	-0	-0.005	-0	-0	0	-3.601
5	Z	-0	-0	0	-0	-0	-0	0	-0	0
6	X	-0	-0	0	0	-0	-0	0	-0	0
6	Y	0.134	-0.252	-0.078	-0.132	-3.561	0.263	-0.174	-0.194	-0
6	Z	-0	0	-0	-0	-0	-0	-0	-0	-6.443
7	X	0	-0	-0	-0	-0	-0	-0	0	-0
7	Y	-1.86	-1.651	1.947	1.829	-0	-1.652	1.805	1.776	-0
7	Z	0	0	0	0	-6.446	0	0	0	0
8	X	0	-0	0	-0	0	0	-0	-0	12.568
8	Y	2.334	2.438	2.332	2.383	0.003	-2.455	-2.406	2.398	0
8	Z	0	0	-0	-0	0	-0	0	-0	0
9	X	-0	-0	-0	0	-0	-0	-0	-0	-0
9	Y	-0.798	-0.781	-0.786	-0.771	-0.156	0.788	-0.76	-0.795	-2.388
9	Z	0	-0	0	0	0	-0	0	-0	-0
10	X	0	0	-0	0	-0	0	-0	-0	-0
10	Y	-1.15	-1.128	1.269	1.232	1.218	-1.148	1.31	1.152	1.006
10	Z	0	-0	0	0	-0	0	0	-0	-0
11	X	0	-0	-0	0	-0	0	0	0	-0

Mode	Direction	ses_QUAD	ses_TRI	aba_QUADR	aba_QUAD	ccx_TRI	ca_TRI	ccx_QUAD	ca_QUAD	ccx_QUADR
11	Y	-0	-0	0	0	-1.9	-0	0	0	0
11	Z	0	-6.451	0	0	0	6.453	0	6.453	3.589

Shell, 2nd order

Table 19: Effective modal mass [kg] per mode, global X/Y/Z — shell, order 2.

Mode	Direction	ses_QUAD	ses_TRI	aba_QUADR	aba_QUAD	aba_TRI	ccx_TRI	ca_TRI	ccx_QUAD	ca_QUAD	ccx_QUADR
1	X	0	0	0	0	0	0	0	0	0	0
1	Y	119.9	119.9	119.9	119.9	119.9	119.9	119.9	119.9	119.9	119.9
1	Z	0	0	0	0	0	0	0	0	0	0
2	X	0	0	0	0	0	0	0	0	0	0
2	Y	0	0	0	0	0	0	0	0	0	0
2	Z	0	0	0	0	0	0	0	0	0	0
3	X	0	0	0	0	0	0	0	0	0	0
3	Y	0	0	0	0	0	0	0	0	0	0
3	Z	122	122	122	122	122	122	122	122	122	122
4	X	0	0	0	0	0	0	0	0	0	0
4	Y	36.9	36.9	36.9	36.9	36.9	36.9	36.9	36.9	36.9	36.9
4	Z	0	0	0	0	0	0	0	0	0	0
5	X	0	0	0	0	0	0	0	0	0	0
5	Y	0	0	0	0	0	0	0	0	0	0
5	Z	0	0	0	0	0	0	0	0	0	0
6	X	0	0	0	0	0	0	0	0	0	0
6	Y	0	0	0	0	0	0	0	0	0	0
6	Z	0	0	0	0	0	0	0	0	0	0
7	X	0	0	0	0	0	0	0	0	0	0
7	Y	3.5	3.4	3.5	3.5	3.5	3.4	3.4	3.4	3.5	3.4
7	Z	0	0	0	0	0	0	0	0	0	0
8	X	0	0	0	0	0	0	0	0	0	0
8	Y	5.5	5.6	5.5	5.5	5.5	5.6	5.5	5.6	5.5	5.5
8	Z	0	0	0	0	0	0	0	0	0	0
9	X	0	0	0	0	0	0	0	0	0	0
9	Y	0.7	0.6	0.7	0.7	0.7	0.6	0.6	0.6	0.7	0.6
9	Z	0	0	0	0	0	0	0	0	0	0
10	X	0	0	0	0	0	0	0	0	0	0
10	Y	1.4	1.4	1.4	1.4	1.4	1.4	1.4	1.4	1.4	1.4
10	Z	0	0	0	0	0	0	0	0	0	0
11	X	0	0	0	0	0	0	0	0	0	0
11	Y	0	0	0	0	0	0	0	0	0	0
11	Z	0	0	0	0	0	0	0	0	0	0

Table 20: Participation factor per mode scaled to unit generalised mass [√kg], global X/Y/Z — shell, order 2.

Mode	Direction	ses_QUAD	ses_TRI	aba_QUADR	aba_QUAD	aba_TRI	ccx_TRI	ca_TRI	ccx_QUAD	ca_QUAD	ccx_QUADR
1	X	-0	-0	0	0	0	-0	0	0	-0	-0
1	Y	-10.952	10.952	10.951	10.951	10.952	10.952	10.952	10.951	10.952	10.951
1	Z	-0	-0	0	0	0	0	-0	0	0	0
2	X	-0	0	0	0	0	-0	-0	0	0	0
2	Y	0	0	0	0	0	0	-0	0	-0	0
2	Z	0	0	0	0	-0	-0	0	-0	-0	-0
3	X	-0	-0	0	0	0	-0	0	0	0	0
3	Y	-0	-0	0	0	0	0	0	0	0	0
3	Z	11.047	-11.047	11.044	11.044	11.047	-11.046	11.047	-11.046	11.047	-11.046
4	X	-0	-0	-0	-0	0	-0	-0	-0	0	0
4	Y	-6.071	6.072	-6.071	-6.071	-6.073	6.072	6.071	6.073	-6.071	6.071
4	Z	0	-0	0	0	0	-0	-0	-0	0	-0
5	X	0	-0	-0	-0	-0	0	-0	-0	0	-0
5	Y	0	0	-0	-0	-0	0	-0	-0	-0	-0
5	Z	-0	0	-0	-0	0	-0	-0	-0	-0	0
6	X	-0	0	-0	-0	-0	-0	0	-0	0	0
6	Y	-0.137	-0.144	-0.137	-0.137	-0.138	-0.144	0.142	0.141	-0.137	0.138
6	Z	0	0	0	0	-0	-0	-0	0	0	0
7	X	0	-0	-0	-0	-0	-0	-0	-0	-0	-0
7	Y	1.859	-1.85	1.858	1.858	1.88	1.848	-1.849	1.839	-1.859	1.852
7	Z	-0	0	-0	-0	0	0	0	0	0	0
8	X	-0	0	-0	-0	-0	0	-0	-0	-0	0

Mode	Direction	ses_QUAD	ses_TRI	aba_QUADR	aba_QUAD	aba_TRI	ccx_TRI	ca_TRI	ccx_QUAD	ca_QUAD	ccx_QUADR
8	Y	-2.35	-2.358	2.351	2.351	2.349	2.356	2.353	-2.362	2.35	-2.35
8	Z	-0	-0	0	0	0	-0	0	0	0	0
9	X	0	0	-0	-0	0	-0	-0	0	-0	0
9	Y	-0.809	-0.801	-0.808	-0.808	-0.818	0.793	0.801	0.781	-0.809	0.799
9	Z	0	0	0	0	-0	-0	0	-0	-0	-0
10	X	-0	-0	-0	-0	-0	-0	0	-0	-0	-0
10	Y	-1.171	-1.185	1.17	1.17	1.175	1.193	1.178	-1.201	1.17	1.178
10	Z	-0	-0	-0	-0	0.001	0	0	-0	0	0
11	X	-0	-0	0	0	0	-0	-0	-0	-0	-0
11	Y	-0	0.001	0	0	0	0.001	-0.001	0	0	0
11	Z	0	-0	-0	-0	0	-0	-0	0	-0	-0

Line, 1st order

Table 21: Effective modal mass [kg] per mode, global X/Y/Z — line, order 1.

Mode	Direction	ses	aba	ccx	ca	ccxR
1	X	0	0	0	0	0
1	Y	0	0	116.4	0	116.4
1	Z	0	0	0	0	0
2	X	0	0	0	0	0
2	Y	116.7	116.6	0	116.5	0
2	Z	0	0	114.8	0	114.8
3	X	0	0	0	0	0
3	Y	0	0	35.7	0	35.7
3	Z	0	0	0	0	0
4	X	0	0	0	0	0
4	Y	0	0	12.3	0	12.3
4	Z	120.2	118.2	0	116.5	0
5	X	0	0	0	0	0
5	Y	0	0	0	0	0
5	Z	0	0	0	0	0
6	X	0	0	0	0	0
6	Y	36.3	0	0	35.8	0
6	Z	0	0	35	0	35
7	X	0	0	153.9	0	153.9
7	Y	0	36.2	0	0	0
7	Z	0	0	0	0	0
8	X	0	0	0	0	0
8	Y	0	0	6.2	0	6.2
8	Z	0	0	0	0	0
9	X	0	0	0	0	0
9	Y	0	0	3.8	0	3.8
9	Z	0	0	0	0	0
10	X	0	0	0	0	0
10	Y	0	0	0	0	0
10	Z	0	0	0	0	0
11	X	0	0	0	0	0
11	Y	0	0	0	0	0
11	Z	0	0	11.4	0	11.4

Table 22: Participation factor per mode scaled to unit generalised mass [√kg], global X/Y/Z — line, order 1.

Mode	Direction	ses	aba	ccx	ca	ccxR
1	X	0	-0	0	-0	0
1	Y	-0	-0	-10.789	0	-10.789
1	Z	-0	0	0	0	0
2	X	0	-0	0	0	0
2	Y	10.803	10.797	-0	10.793	-0
2	Z	-0	-0	-10.716	0	-10.716
3	X	0	0	-0	-0	-0
3	Y	-0	-0	5.979	-0	5.979
3	Z	-0	0	-0	-0	-0
4	X	0	0	0	-0	0
4	Y	-0	-0	3.5	-0	3.5
4	Z	-10.963	10.87	-0	10.793	-0
5	X	-0	-0	0	0	0

Mode	Direction	ses	aba	ccx	ca	ccxR
5	Y	-0	0	-0	0	-0
5	Z	0	0	0	0	0
6	X	0	-0	0	-0	0
6	Y	-6.024	-0	0	-5.981	0
6	Z	0	0	-5.913	-0	-5.913
7	X	-0	0	12.407	-0	12.407
7	Y	-0	-6.016	-0	-0	-0
7	Z	-0	-0	0	-0	0
8	X	0	-0	0	-0	0
8	Y	-0	-0	2.498	-0	2.498
8	Z	-0	0	0	0	0
9	X	-0	0	0	-0	0
9	Y	0	0	-1.937	0	-1.937
9	Z	0	0	0	-0	0
10	X	-0	0	-0	-0	-0
10	Y	-0	0	0	0	0
10	Z	-0	-0	-0	0	-0
11	X	-0	0	0	-0	0
11	Y	-0	-0	0	0	0
11	Z	-0	0	-3.374	0	-3.374

Line, 2nd order

Table 23: Effective modal mass [kg] per mode, global X/Y/Z — line, order 2.

Mode	Direction	ses	aba
1	X	0	0
1	Y	0	0
1	Z	0	0
2	X	0	0
2	Y	176.8	116.6
2	Z	0	0
3	X	0	0
3	Y	0	0
3	Z	0	0
4	X	0	0
4	Y	0	0
4	Z	180	118.2
5	X	0	0
5	Y	0	0
5	Z	0	0
6	X	0	0
6	Y	55	0
6	Z	0	0
7	X	0	0
7	Y	0	36.2
7	Z	0	0
8	X	0	0
8	Y	0	0
8	Z	0	0
9	X	0	0
9	Y	0	0
9	Z	0	0
10	X	0	0
10	Y	0	0
10	Z	0	0
11	X	0	0
11	Y	0	0
11	Z	0	0

Table 24: Participation factor per mode scaled to unit generalised mass [√kg], global X/Y/Z — line, order 2.

Mode	Direction	ses	aba
1	X	0	0
1	Y	0	0
1	Z	-0	-0
2	X	-0	-0

Mode	Direction	ses	aba
2	Y	13.297	10.798
2	Z	0	-0
3	X	-0	0
3	Y	-0	-0
3	Z	-0	-0
4	X	0	-0
4	Y	-0	-0
4	Z	-13.418	10.871
5	X	0	-0
5	Y	0	0
5	Z	-0	0
6	X	0	-0
6	Y	-7.413	0
6	Z	0	-0
7	X	-0	0
7	Y	-0	-6.015
7	Z	-0	-0
8	X	-0	0
8	Y	-0	0
8	Z	-0	-0
9	X	0	0
9	Y	0	0
9	Z	0	-0
10	X	0	-0
10	Y	0	0
10	Z	-0	0
11	X	0	-0
11	Y	0	-0
11	Z	-0	0

Cantilever eigenvalue analysis

Line, 1st order

Figures unavailable for: Abaqus, Calculix, Code_Aster, Sesam.

Line, 1st order, reduced integration

Figures unavailable for: Calculix.

Line, 2nd order

Figures unavailable for: Abaqus, Sesam.

Shell, 1st order, TRI

Figures unavailable for: Calculix, Code_Aster, Sesam.

Shell, 1st order, QUAD

Figures unavailable for: Abaqus, Calculix, Code_Aster, Sesam.

Shell, 1st order, QUAD, reduced integration

Figures unavailable for: Abaqus, Calculix.

Shell, 2nd order, TRI

Figures unavailable for: Abaqus, Calculix, Code_Aster, Sesam.

Shell, 2nd order, QUAD

Figures unavailable for: Abaqus, Calculix, Code_Aster, Sesam.

Shell, 2nd order, QUAD, reduced integration

Figures unavailable for: Abaqus, Calculix.

Solid, 1st order, TET

Figures unavailable for: Abaqus, Calculix, Code_Aster, Sesam.

Solid, 1st order, HEX

Figures unavailable for: Abaqus, Calculix, Code_Aster, Sesam.

Solid, 1st order, HEX, reduced integration

Figures unavailable for: Abaqus, Calculix, Code_Aster.

Solid, 2nd order, TET

Figures unavailable for: Abaqus, Calculix, Code_Aster, Sesam.

Solid, 2nd order, HEX

Figures unavailable for: Abaqus, Calculix, Code_Aster, Sesam.

Solid, 2nd order, HEX, reduced integration

Figures unavailable for: Abaqus, Calculix, Code_Aster.

Plate eigenvalue analysis

The plain and the stiffened strip, at the eigen case's 0.0625 m seed.

Shell, 1st order, plain

Figures unavailable for: Abaqus, Calculix, Code_Aster, Sesam.

Shell, 1st order, stiffened

Figures unavailable for: Abaqus, Code_Aster, Sesam.

Plate static analysis

The deflected strip under 1 kPa, plain and stiffened, at the same 0.0625 m seed. The other seeds deform into the same shape; their convergence is in the summary table.

Shell, 1st order, plain, seed 0.0625 m

Figures unavailable for: Abaqus, Calculix, Code_Aster, Sesam.

Shell, 1st order, stiffened, seed 0.0625 m

Figures unavailable for: Abaqus, Code_Aster, Sesam.